

Alekya Kowta

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Product Management | Full-Stack Software Development | Java, Python & REST APIs | AI/Agentic Systems | Agile/Scrum Delivery

Experience

Product Manager, Capital One (via Mindlance)

June 2026 – Aug 2026

Product Owner, Account Servicing

McLean, VA

- Serve as Product Owner for the Account Servicing domain and Product Manager for two engineering teams (10 engineers each), overseeing platforms spanning Cases (Complaints, Claims, Fraud Investigations), document association, customer activity/agent notes, and beneficiary/trust entity management
- Lead product strategy and delivery for the DFS-to-Capital One migration, coordinating across upstream (DFS), internal engineering, and downstream product teams on requirements, sequencing, and decisions at each stage; partner with Legal on sensitive domains including fraud, claims, and trust entities
- Direct data mapping across account servicing systems for migration wave planning, proactively flagging issues prior to execution to reduce downstream risk
- Executed the August migration wave, moving 819,236 accounts from DFS to Capital One; drove remediation of identified issues, improving reliability and reducing migration risk

Software Engineer I / Software Engineering Intern, Dell Technologies

Jan 2022 – Aug 2025

Full-Stack & Backend Development

Bangalore, India

- Designed and built full-stack REST APIs and backend services supporting 150M+ connected devices, working across the complete SDLC from design through deployment in close collaboration with engineering, architecture, and product teams
- Led infrastructure migration to Kubernetes, improving deployment throughput by 20% and cutting deployment latency 35% across microservices through close teamwork with DevOps and platform engineers
- Built and maintained relational and NoSQL data layers supporting 10M+ daily transactions with zero downtime; presented technical decisions directly to engineering leadership
- Standardized Agile/Scrum workflows in Confluence (cutting onboarding time 30% across 3 teams) and built monitoring dashboards tracking 15+ service health metrics (cutting incident detection time 25%); wrote Python automation scripts saving ~4 hours per release cycle

Projects

Debatrium – Multi-Agent AI System – Python, AWS, GPT-4o, NVIDIA NIM, REST APIs, SQS, Redis

- Architected a 9-agent multi-agent AI system orchestrating GPT-4o and NVIDIA NIM models to run structured, multi-perspective debate, with an iterative feedback/quality-control loop refining agent outputs
- Designed and built the distributed backend on AWS using EC2 Auto Scaling Groups, 12 SQS FIFO queues with dead-letter handling, and ElastiCache Redis (Multi-AZ, TLS), secured via API Gateway and Firebase Authentication

Carbon-Aware Freight Routing – Python, Pandas, PyTorch

- Built an end-to-end data pipeline and routing application processing a ~960-node road network with real-world emissions and elevation data; benchmarked multiple algorithmic approaches, improving routing efficiency while reducing estimated carbon output

Security Research & Tooling – Python, Network Security – Published IEEE & Springer

- Built and evaluated reconnaissance tooling for penetration-testing workflows and researched IoT vulnerability classes across device, network, and protocol layers; published in IEEE conference proceedings and co-authored a Springer chapter, *Cyber Intelligence and Information Retrieval*

Trustworthy AI Hackathon – 1st Place Winner – Python, Pandas, Jupyter

- Built an open-source analytical tool translating complex datasets into clear summary statistics and visualizations for non-technical stakeholders, demonstrating strong problem-solving and communication under hackathon time constraints

Technical Skills

Languages: Java, Python, C#, JavaScript, SQL, C++

Full-Stack & Web: HTML, CSS, React, Spring Boot, ASP.NET Core MVC, REST APIs, Microservices, .NET

Product & Agile: Product Strategy, Roadmap Planning, Stakeholder Management, Agile/Scrum, Sprint Planning, Backlog Management (Jira), Requirements Gathering, TDD, Unit & Integration Testing, Code Review, Git

Cloud & DevOps: AWS (EC2, Lambda, S3), Microsoft Azure, Docker, Kubernetes, CI/CD, Jenkins, Linux

Data & Tools: Oracle, MySQL, NoSQL, Redis, Kafka, PowerBI, Grafana, Jira, Confluence

AI/ML (Supplemental): OpenAI GPT-4o API, Prompt Engineering, PyTorch, Pandas

Education

The George Washington University

Washington, DC

M.S. in Computer Science, GPA: 3.9/4.0 (*Data Structures and Algorithms, ML and AI, Distributed Systems*)

May 2027

Vellore Institute of Technology (VIT)

Tamil Nadu, India

B.Tech in Information Technology, GPA: 9.04/10

May 2022